

Ismam Nur Swapnil

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EDUCATION

Bangladesh University of Engineering and Technology

BSc in Electrical and Electronic Engineering

CGPA: 3.73/4.0

Feb 2020 – Feb 2025

Dhaka, Bangladesh

Selected coursework: Calculus and Differential Equations; Linear Algebra; Probability and Statistics; Numerical Methods; Signals and Linear Systems; Digital Signal Processing; Control Systems; Artificial Intelligence and Machine Learning; Robotics and Automation; Digital Image Processing.

Thesis: AI-driven dermatological diagnosis using vision-language models; SFT, QLoRA, and DPO. Thesis poster

RESEARCH FOCUS

I study how agents can learn, perceive, and reason with less data and computation. My research spans reinforcement learning, world models, visual reasoning, and on-policy distillation, with computational and data efficiency as a unifying goal.

PUBLICATIONS & PREPRINTS

[1] Skip What You Can Predict: Predictive Repositioning for Policy Optimization for Efficient LLM Training

I. N. Swapnil, A. Saha, T. Zahra, T. A. Khan, M. A. Haque, & S.-N. Lim

Rather than executing every same-batch optimizer step in RLVR, extrapolate from two observed steps to a farther point on the trajectory and correct; matches performance in fewer steps and less wall-clock time, with error bounds for AdamW and Muon.

Submitted to ICLR 2027 · Preprint · First author · arXiv:2605.06755

[2] Teach Yourself Where to Look: On-Policy Attention Self-Distillation for Reasoning

S. H. Arib, R. Akter, I. N. Swapnil, Md. F. A. Sayeedi, T. Mohiuddin*, & Md M. Islam* (*equal supervision)

Distills not only *what* the teacher predicts but *where it looks*, by projecting teacher attention onto student-visible context;

+4.98–8.40 points over token-only distillation on Qwen3 (1.7B–8B), with 73.9% fewer rollout tokens and 1.53× faster training.

Submitted to ICLR 2027 · Preprint · arXiv:2609.33200

[3] GRPO++: Enhancing Dermatological Reasoning under Low Resource Settings

I. N. Swapnil, A. Saha, T. A. Khan, & M. A. Haque

Stabilizes data-hungry GRPO for low-resource medical VLMs; a GRPO++ → SFT → knowledge-graph DPO pipeline adds diagnostic reasoning while reducing factual errors.

Under review at IEEE Journal of Biomedical and Health Informatics · First author · arXiv:2510.01236

[4] CLARIFY: A Specialist-Generalist Framework for Accurate and Lightweight Dermatological Visual Question Answering

A. Saha, T. A. Khan, I. N. Swapnil, & M. A. Haque

A small specialist classifier guides a compressed conversational VLM, grounded by knowledge-graph retrieval; +18% diagnostic accuracy over a fine-tuned full-size VLM with ≥20% less VRAM.

Preprint · arXiv:2508.18430

RESEARCH EXPERIENCE

University of Central Florida

Research Collaborator

Nov 2025 – Present

Orlando, FL (Remote)

- Collaborating with **Prof. Ser-Nam Lim** on efficient policy optimization for large language model reasoning.
- **First author of PrePO:** predictive repositioning approximates repeated same-batch optimizer updates at a fixed active cost, reducing explicit optimization steps and wall-clock time at matched performance.
- Established **finite-horizon error bounds for AdamW and Muon**; validated on RL with verifiable rewards for LLM reasoning, with gains carrying over to supervised fine-tuning.

Collaboration with Md Mofijul Islam (Amazon GenAI)

Research Collaborator, On-Policy Self-Distillation

2026

Remote

- Co-developed **OPASD**, which adds solution-conditioned **attention supervision** to on-policy self-distillation; on Qwen3 (1.7B–8B) it gains 4.98–8.40 points over token-only OPSD on four competition-level math benchmarks with 72.6% less compute.

Independent Research

Feb 2025 – Present

World Models and On-Policy Distillation

- Working on **world models** for reinforcement learning and **on-policy distillation** for language model reasoning.
- Investigating sampling strategies and inference-time scaling to improve the efficiency of language model reasoning.

Bangladesh University of Engineering and Technology

Feb 2024 – Feb 2025

Undergraduate Researcher

Dhaka, Bangladesh

- Developed **GRPO++**, a stabilized GRPO for data-scarce medical VLMs, within a three-stage pipeline: GRPO++ for diagnostic reasoning, SFT for conversation, and knowledge-graph DPO to reduce factual errors.
- Designed **CLARIFY**, a specialist-generalist architecture for dermatological VQA: +18% diagnostic accuracy over a fine-tuned full-size VLM with $\geq 20\%$ less VRAM.

PROFESSIONAL EXPERIENCE

Advanced Chemical Industries Limited

Jun 2025 – Present

Machine Learning Engineer, Department of MIS

Dhaka, Bangladesh

- **Primary focus:** developing a **frontier OCR model** for Bangla handwritten and printed documents.
- Built **AgroResearch AI**, a retrieval-augmented system for synthesizing agricultural research.
- Developing **Bangla text-to-speech** for natural and expressive audio synthesis.

SELECTED PROJECTS

- **Bangla OCR Pipeline** Code
Combined YOLO word detection with Qwen3-VL-2B recognition for handwritten and printed Bangla; implemented PyTorch and ONNX deployment backends.
- **AgroResearch AI** Code
Built a retrieval-augmented chatbot that synthesizes scientific literature for crop, soil, livestock, and aquaculture management.
- **Real-Time AI Sales Communicator** Code
Built a full-duplex voice agent with a local Qwen LLM, Whisper speech recognition, and pyttsx3 speech synthesis, served through FastAPI and WebSockets.

TECHNICAL SKILLS

Methods	Policy optimization, on-policy distillation, multimodal LLMs, RLHF, SFT, QLoRA, DPO, computer vision, RAG
Programming	Python, MATLAB
ML frameworks	PyTorch, TensorFlow, Keras, Scikit-learn
Agents & tools	LangChain, LangGraph, LlamaIndex, FastAPI, Docker, Git
Data processing	Pandas, NumPy, OpenCV, SciPy

HONORS & ACADEMIC ACTIVITIES

- **University Merit Scholarship**, BUET; awarded for academic excellence in two semesters.
- **Dean's List Award**, BUET; recognized for a GPA of 3.75 or above.
- **139th place**, BUET undergraduate admission test, among over 12,000 candidates.
- **Runner-up, Poster Presentation**, Bear Summit 2024, Department of EEE, BUET; vision-language models for dermatological diagnosis. [Award](#)
- **Participant**, National Semiconductor Symposium 2024. [Certificate](#)

REFERENCES

Dr. Mohammad Ariful Haque

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Dr. Mohammad Jahangir Alam

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